

$$\max_{\theta} \log p_{\theta}(x) \rightarrow \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$

$$\log p_{\theta}(x) = \int q_{\phi}(z|x) \log p_{\theta}(x) dz = \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x)]$$

이때

$$p_{\theta}(x) = \frac{p_{\theta}(x, z)}{p_{\theta}(z|x)} = \frac{p_{\theta}(x|z) p(z)}{p_{\theta}(z|x)}$$

$$\begin{aligned} \Rightarrow \log p_{\theta}(x) &= \log p_{\theta}(x|z) + \log \frac{p(z)}{p_{\theta}(z|x)} \\ &= \log p_{\theta}(x|z) + \log \frac{p(z)}{q_{\phi}(z|x)} + \log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \end{aligned}$$

따라서

$$\begin{aligned} \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x)] &= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{p(z)}{q_{\phi}(z|x)} \right] \\ &\quad + \mathbb{E}_{q_{\phi}(z|x)} \left[\log \frac{q_{\phi}(z|x)}{p_{\theta}(z|x)} \right] \\ &= \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z)) \\ &\quad + \text{KL}(q_{\phi}(z|x) \| p_{\theta}(z|x)) \\ &\geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z)) \end{aligned}$$

$$\therefore \log p_{\theta}(x) \geq \mathbb{E}_{q_{\phi}(z|x)} [\log p_{\theta}(x|z)] - \text{KL}(q_{\phi}(z|x) \| p(z))$$